

THERMACART · DESIGN PRECIS

ThermaCart design precis

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	TCT-PRC-001	Version	0.2
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/thermacart		
Source	docs/02-concept.md @ 66b9511		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, grades, first-order numbers, safety, media)
0.1	2026-09-25	Amish Chadha	Initial scaffold

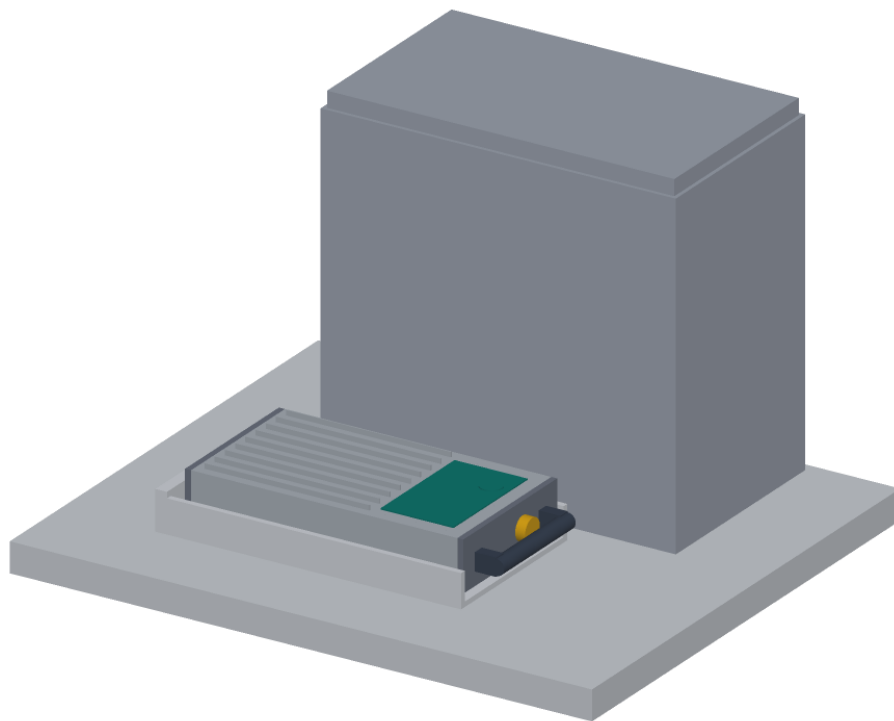
ThermaCart design precis

■ Summary

ThermaCart is a sealed aluminium cartridge, 322 x 152 x 61 mm, that holds about 1.1 kg of phase-change material (PCM) in one of three standard grades: C5 (melts at 5 °C, cold chain), C25 (25 °C, comfort) and H70 (70 °C, hot holding). It fits a GN 1/3 gastronorm slot, carries by a handle, shows its state through a sight window and locks into an adapter frame whose key accepts only the right grade. First-order estimates give 55 to 67 Wh of usable storage per cartridge, 2.6 kg filled and about \$49 in parts. Two requirements are not met on paper (R5, PCM mass fraction; R14, hot handle on H70), and two are at risk or not met for the C5 grade (R3, R10). All numbers are estimates, and every design choice below is proposed, awaiting Amish.

ThermaCart

CONCEPT, NOT FOR FABRICATION



Grey: Table top, 25 L cooler for scale

Figure 1. A TC-L cartridge (C5 grade) in its adapter frame on a table, with a 25 L cooler for scale.

■ How it works

1. **Charge.** The cartridge is put where the right temperature already exists: a freezer or cold room for C5, a refrigerator or night air for C25, an oven at 100 °C or less, a heating pad or a heat store for H70. The PCM freezes (C5, C25) or melts (H70) and stores latent heat at a nearly constant temperature.

2. **Carry and swap.** The sight window shows the state. The owner carries the cartridge to a cooler, food carrier, cabinet or heat exchanger and slides it into an adapter frame.
3. **Hold.** The host box leaks heat in (cold grades) or out (H70). The PCM absorbs or releases it while staying near its melt point, so the box contents stay in band until the PCM has fully changed phase.
4. **Return.** The discharged cartridge goes back to the charging point. Nothing is consumed; the fill is sealed for life.

ThermaCart: energy per charge cycle, one TC-L H70 cartridge for hot holding (all values are estimates)

CONCEPT, NOT FOR FABRICATION

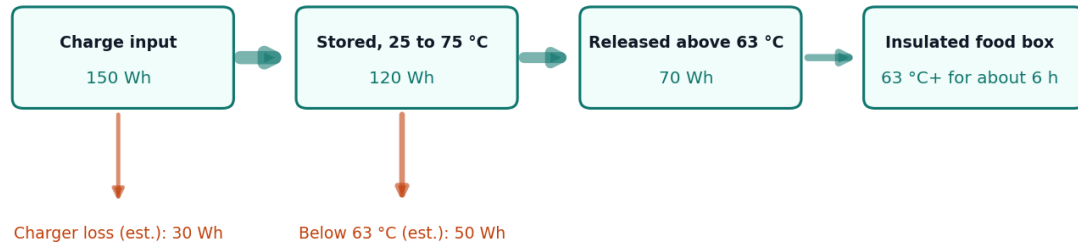


Figure 2. Energy per charge cycle for one H70 cartridge used for hot holding. All values are estimates: about 150 Wh in, 120 Wh stored between 25 and 75 °C, and about 70 Wh released above 63 °C. The remainder is released below 63 °C and is useful only for warming.

■ Standard grades

Table 1. Proposed grades (proposed, awaiting Amish)

Grade	Melt point	Proposed fill (class)	Usable storage per TC-L (estimate)	Typical hosts	Colour and key
C5	5 °C	Organic paraffin, Rubitherm RT 5 HC class	About 55 Wh (198 kJ)	Produce and dairy coolers, a ColdPod-type box after qualification	Blue, key left
C25	25 °C	Organic paraffin, RT 25 HC class	About 61 Wh (220 kJ)	Electronics and battery cabinets, comfort cooling	Green, key centre
H70	70 °C	Organic paraffin, RT 70 HC class	About 67 Wh (242 kJ)	Insulated GN food carriers, keep-warm boxes	Red, key right
W0 (option)	0 °C	Water with a nucleating agent	About 133 Wh (478 kJ)	Produce only; never for vaccines or medicines	White, key outer left

Paraffins are proposed for the first three grades because they are chemically stable, non-corrosive to aluminium and cycle well ([PATH PCM study](#)). Salt hydrates would store roughly 1.5 to 2 times more per litre (estimate), but they supercool, separate and corrode aluminium ([corrosion study](#)), so they are left for a later lined variant.

■ Main components

Table 2. Main components. Numbers match the exploded view (Figure 3) and `bom/bom.csv`.

#	Component	Proposed choice	Notes
1	Cartridge shell	6063 aluminium rectangular tube, 152.4 x 50.8 x 3.18 mm (6 x 2 x 1/8 in), 270 mm long, with nine 5 mm fin strips bonded top and bottom	Stock section, no welding; fins double the air-side area
2	PCM fill	About 1.1 kg of the grade's PCM, filled hot to 90 % of the inner volume	Ullage for expansion; see Safety
3	End caps (2)	12 mm aluminium plate with a 10 mm plug, FKM O-ring and six countersunk screws each	FKM (or NBR) resists paraffin; EPDM does not
4	Fill port plug	G 1/2 or G 3/4 plug with a bonded FKM seal in the handle-end cap	Fill, drain and recycling
5	Handle and keyed nose	Bent aluminium handle at one end with an insulating grip sleeve (for R14); key tab on the other end whose position codes the grade	Key positions in Table 1
6	Grade label and melt indicator	Colour-coded polyester label with grade, fill, melt point and warnings; sight window over a small glass vial of the same PCM	Paraffin is opaque white when solid and clear when liquid
7	Cabinet adapter frame	Bent 1.5 mm aluminium sheet tray with side rails and a keyed stop	One per host slot; host designs copy its key

ThermaCart: exploded view

CONCEPT, NOT FOR FABRICATION

- 1 Cartridge shell, finned aluminium tube
- 2 PCM fill, C5 grade (5 °C)
- 3 End caps with O-ring seals (2)
- 4 Fill port plug and seal
- 5 Handle and keyed nose
- 6 Grade label and melt indicator
- 7 Cabinet adapter frame

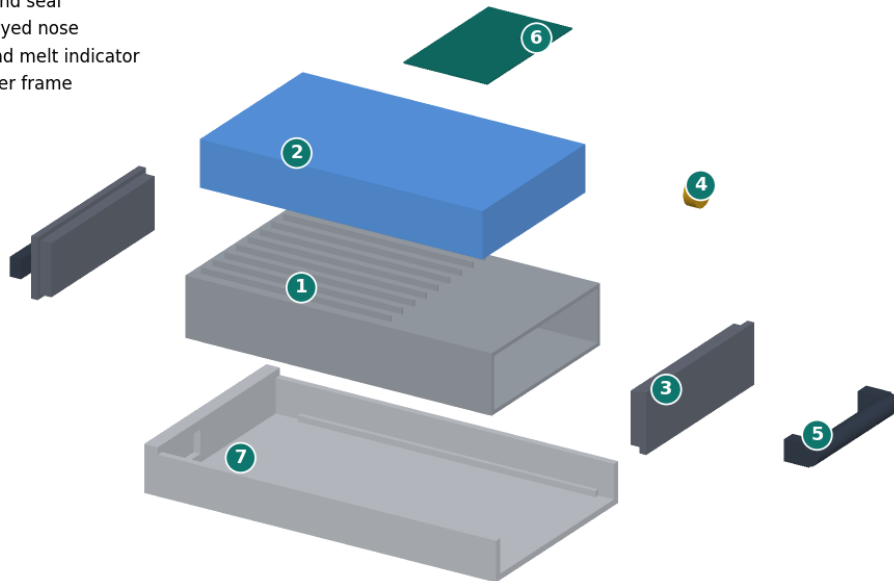


Figure 3. Exploded view. 1 shell, 2 PCM fill, 3 end caps, 4 fill port plug, 5 handle and keyed nose, 6 label and indicator, 7 adapter frame.

ThermaCart: cutaway

CONCEPT, NOT FOR FABRICATION

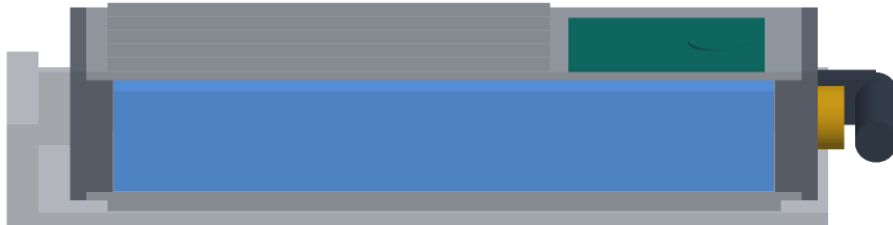


Figure 4. Section along the cartridge. The PCM (blue) fills about 90 % of the tube height; the top 10 % is ullage. The end-cap plugs, fins and label band are visible.

■ First-order numbers

All values are estimates for concept review and will be checked by calculation at TRL 3.

Table 3. First-order numbers

Quantity	Estimate	Basis and assumptions	Requirement
Overall size	322 x 152 x 61 mm	Massing model; 270 mm tube, 6 mm cap flanges, 22 mm handle, 10 mm key, 5 mm fins	R2 met (GN 1/3: 325 x 176 mm)
Inner volume	About 1.62 L	146 x 44.4 mm inside the tube, 250 mm between cap plugs	
PCM mass	About 1.1 kg	90 % fill at about 0.77 kg/L liquid density	
Usable storage	C5 55 Wh, C25 61 Wh, H70 67 Wh	180, 200 and 220 kJ/kg within ± 3 K, below Rubitherm's quoted 250, 230 and 260 kJ/kg (Rubitherm)	R3 not met for C5
Shell and fittings	About 1.5 kg	Aluminium at 2.7 kg/L; caps pocketed to about 60 cm ³ each	
Filled mass	About 2.6 kg	Shell plus fill	R4 met; R5 not met (about 42 % PCM)
Cold hold, two C5 in a 25 L cooler	About 8.1 h at 32 °C	110 Wh over a 13.5 W leak (0.5 W/K x 27 K, estimate)	R10 at risk, no margin
Hot hold, one H70 in a GN carrier	About 6 h at 25 °C	About 70 Wh above 63 °C over an 11 W leak (0.25 W/K x 45 K, estimate)	R11 met
C5 recharge in a –18 °C freezer	About 4 to 8 h	Solidification of an 18 mm half-thickness of paraffin (k about 0.2 W/(m·K)) plus fin convection, first-order Stefan estimate	R12 at risk
H70 recharge on a 60 W pad	About 2.5 h	About 120 Wh stored at about 80 % transfer efficiency	R12 met
Energy density	About 21 to 26 Wh/kg	Usable storage over filled mass	
Parts cost	About \$49 per cartridge, \$15 per frame	Indicative prices, bom/bom.csv	R16 met, thin margin

■ Key design choices (all proposed, awaiting Amish)

- Envelope: GN 1/3 footprint from a stock tube.** Options were (a) a 6 x 2 in stock aluminium tube cut to fit a GN 1/3 slot; (b) the WHO PQS 0.6 L pack envelope, 190 x 120 x 34 mm ([WHO PQS E005](#)), which fits existing vaccine carriers but holds only about 0.5 kg; (c) a custom extruded finned profile. Recommendation: (a) for the first build, with (b) as a later small size (TC-S) and (c) only if volumes justify a die.
- Three paraffin grades first (C5, C25, H70), water as an option.** See Table 1. Recommendation: build C5 first because it has the lowest hazard and links to ColdPod.
- Mechanical grade keying.** A nose tab whose position depends on the grade, and a matching stop in each frame, so a frozen water or hot H70 cartridge cannot be put into a cold-chain box. Recommendation: adopt, and publish the key table as part of the open standard.
- Passive indicator.** A sight window over a vial of the same PCM shows solid or liquid without electronics. Alternative: a thermochromic strip (cheaper, reads surface temperature rather than state of charge). Recommendation: sight window.

5. **Fill hot, seal for life.** Filling as liquid at the top of the service temperature fixes the ullage at its smallest, so the PCM can only shrink afterwards and cannot pressurise the shell.
6. **Aluminium shell for paraffins, no HDPE.** Paraffin can swell and permeate polyethylene ([compatibility study](#)); aluminium is compatible and conducts heat to the fins. A lined or HDPE variant is kept for salt hydrates and water.

■ Safety

Safety: Paraffin fills are combustible. Never charge an H70 cartridge on an open flame, a gas ring or a hot plate, never above 100 °C, and keep all grades away from fire. If a cartridge leaks, stop using it; spilled liquid paraffin is slippery and burns.

Safety: H70 cartridges reach about 75 °C and can burn skin. The handle must have an insulating grip (R14 is not met with bare aluminium), the label must carry a hot-surface warning, and hot cartridges should only be moved with the handle.

Safety: Do not fill, refill or open a cartridge that is warm or under pressure. Fill only as described in the fill recipe, with the specified ullage. Never seal a cartridge that is only partly melted or at room temperature for an H70 grade, as later heating could pressurise it.

Safety: A C5 cartridge charged in a freezer is below 0 °C when it comes out. Condition it until the sight window shows the first liquid before using it next to anything that must not freeze. ThermaCart is not qualified for vaccines or medicines, and the W0 water grade must never be used with them.

- Aluminium edges and fins are sharp after cutting: deburr all edges.
- A 2.6 kg cartridge dropped on a foot can injure: keep cartridges low in racks.

■ Open questions for TRL 3

- Can the C5 grade reach R3 and R10? Options: a thinner wall, a slightly taller tube within the GN depth, or a denser PCM.
- R5: is a 42 % PCM fraction acceptable for the first build, with a custom extrusion later?
- Charging C5 in a domestic freezer overcools it; is a conditioning step acceptable, or should C5 be charged in a refrigerator at 0 to 2 °C?
- Seal design and O-ring material per grade, and the drop case for a filled cartridge (R6).
- Handle grip material for H70 and the handle temperature (R14).
- Which host first: a produce cooler with a trader partner, a GN food carrier with a caterer, or ColdPod?
- Does ThermaCart become a declared dependency of ColdPod or ZeerBox? Any change to those repos is Amish's decision.

Concept media: [blueprint sheet](#), [interactive 3D model](#).